

Sentiment Analysis of Movie Reviews Using Support Vector Machines

A Comparative Study of Linear, RBF, and Polynomial SVM Kernels

Atanu Roy

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Abstract

Sentiment analysis of textual reviews is a widely studied problem in Natural Language Processing (NLP) with significant practical applications. In this work, I address sentiment classification of movie reviews as a binary text classification task using Support Vector Machine (SVM)-based approaches. I conduct a systematic empirical comparison of three SVM kernel variants — Linear, Radial Basis Function (RBF), and Polynomial — on the Stanford Large Movie Review Dataset (IMDB), comprising 50,000 labeled reviews. Text preprocessing, TF-IDF feature extraction with unigram and bigram representations, and extensive hyperparameter tuning are employed. Experimental results demonstrate that the Linear SVM significantly outperforms both the RBF and Polynomial kernels across all evaluation metrics, achieving an accuracy of 88.72%, an F1-score of 0.887, and an ROC-AUC of 0.956 at the optimal regularization parameter $C = 0.1$. The failure of nonlinear kernels on high-dimensional sparse TF-IDF representations is analyzed and discussed in detail. A misclassification analysis is also presented to identify the linguistic phenomena responsible for model errors. Furthermore, model stability is validated through 5-fold cross-validation, and a learning curve analysis with overfitting remediation experiments is conducted to characterize the generalization behavior of the best-performing model.

1. Introduction

Online reviews contain valuable opinions that exert considerable influence on consumer decisions. However, manually analyzing thousands of reviews at scale is both time-consuming and impractical. Automated machine learning models capable of classifying text-based reviews as positive or negative based on their underlying sentiment are therefore of significant practical and research interest.

In this work, I address sentiment analysis as a binary text classification problem using Support Vector Machine (SVM)-based approaches. The study not only focuses on building an effective sentiment classifier but also on systematically comparing different SVM kernels — namely, Linear SVM, SVM with Radial Basis Function (RBF) kernel, and SVM with Polynomial kernel — to understand and analyze their behavior on high-dimensional text data.

For this study, I employ the Stanford Large Movie Review Dataset (IMDB) [1], a widely used

benchmark dataset containing 50,000 reviews evenly split into training and test sets. My primary objectives are:

- To train an accurate binary sentiment classifier on the IMDB dataset.
- To empirically compare three SVM kernel variants under controlled experimental conditions.
- To understand why certain kernels succeed or fail on high-dimensional sparse text representations.
- To analyze the linguistic characteristics of misclassified samples to identify the fundamental limitations of the approach.
- To validate model stability through cross-validation and analyze generalization behavior via learning curve experiments, including systematic overfitting remediation attempts.

2. Methodology

The overall methodology follows a standard machine learning pipeline consisting of data ingestion, text preprocessing, feature extraction, model training, hyperparameter tuning, evaluation, and error analysis. The complete workflow is illustrated in Figure 1.

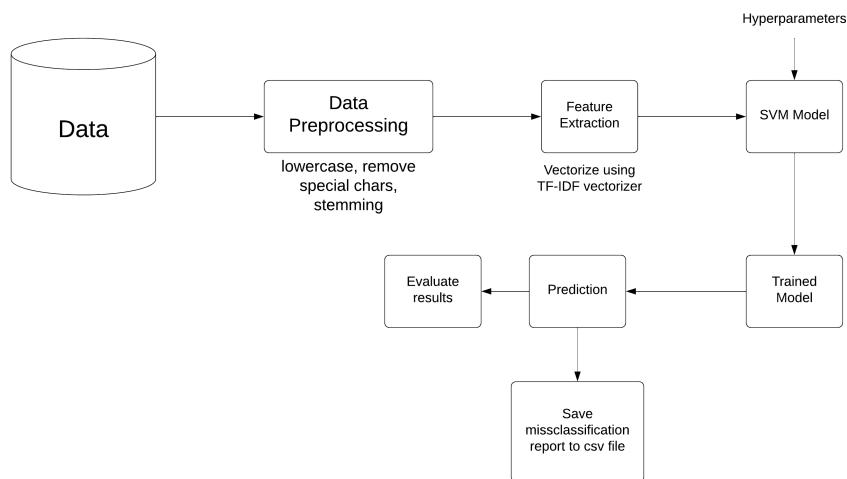
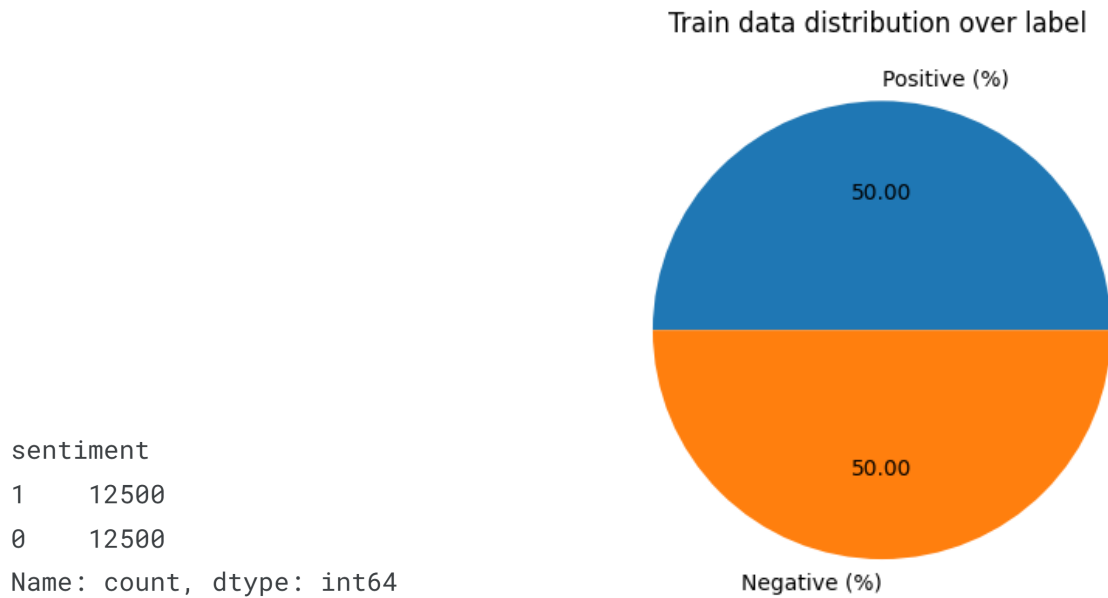


Figure 1: Methodology flowchart illustrating the end-to-end pipeline from raw data to misclassification report.

2.1 Dataset Description

I use the Stanford Large Movie Review Dataset (IMDB) [1], which consists of 50,000 English movie reviews labeled as either positive or negative. The dataset is evenly balanced, comprising 25,000 positive and 25,000 negative reviews, and is pre-split into 25,000 training and 25,000 testing samples. This dataset is widely adopted as a standard benchmark for sentiment analysis research.

Positive reviews are assigned numerical label 1 and negative reviews are assigned label 0. The training set distribution is perfectly balanced, as shown in Figure 2.



(a) Sentiment label counts in training data.

(b) Training data class distribution as a pie chart.

Figure 2: Training data class distribution. Both classes account for exactly 50% of the training set (12,500 samples each).

Due to the substantially higher computational cost of kernel-based SVMs, the full 25,000 training samples were used only for the Linear SVM. For SVM models with RBF and Polynomial kernels, a reduced subset of 5,000 training samples was used to maintain experimental feasibility.

2.2 Text Preprocessing

Before feature extraction and model training, all raw review texts are preprocessed to reduce noise and standardize linguistic form. The preprocessing pipeline consists of the following steps:

1. **Lowercasing:** All characters are converted to lowercase to ensure case-insensitive token matching.
2. **Removal of special characters:** HTML tags, punctuation, and other non-alphanumeric characters are removed.
3. **Stemming:** Words are reduced to their root forms to ensure that morphological variants of the same word are treated uniformly.

Preprocessing ensures that semantically identical words (e.g., “Running,” “running,” “runs”) are mapped to the same token, thereby improving model generalization. Example outputs of the preprocessing pipeline are shown in Figures 3 and 4.

I've Seen The Beginning Of The Muppet Movie, But Just The Half. Because I Only Watched It At Mrs Kelly's Friend's House. The Songs Were The Best And The Muppets Were So Hilarious. They Learn That If They Believe In The End Of The Rainbow, Anyone Can Make It, No Matter How Small, No Matter How Green(Which Was Included In The Trailer).

Kermit Is My Favorite Protagonist(Which Means It Describes The Main Character) And So Are The Other Muppets. Mel Brooks Was Amazing When He Played Professor Max Krassman. The Scene Where Miss Piggy Saves Kermit By Doing Kung Fu On Those Guys. It Was So Cool.

The Muppet Movie Is The Best Jim Henson Film With The Most Hilarious Characters And People Will Cherish For His Successful Film.

Figure 3: A sample review before text preprocessing.

ive seen begin muppet movi half watch mr kelli friend hous song best muppet hilari learn believ end rainbow anyon make matter small matter greenwhich includ trailerkermit favorit protagoni stwhich mean describ main charact muppet mel brook amaz play professor max krassman scene miss piggi save kermit kung fu guy coolth muppet movi best jim henson film hilari charact peopl cherish success film

Figure 4: The same review after text preprocessing.

2.3 Feature Extraction

Since machine learning models operate on numerical data, textual reviews must be transformed into numerical feature vectors. For this purpose, I employ Term Frequency–Inverse Document Frequency (TF-IDF) vectorization.

TF-IDF assigns higher weights to terms that are frequent within a given review but rare across the entire corpus, making it particularly effective for capturing sentiment-bearing vocabulary. Both unigrams and bigrams are used to capture local linguistic context, including negation patterns (e.g., “not good,” “very bad”). Trigrams are excluded, as they are too rare to contribute statistically meaningful features. Sublinear term frequency scaling is applied to reduce the disproportionate influence of very frequent words in longer reviews, using the transformation:

$$TF_{\log} = 1 + \log(TF) \quad (1)$$

Two TF-IDF configurations are employed, one for the Linear SVM and a smaller one for the kernel-based SVMs, as detailed below.

2.3.1 TF-IDF Configuration for Linear SVM

- **Maximum vocabulary size:** 25,000 features, balancing training speed with lexical coverage.
- **N-gram range:** Unigrams and bigrams (1, 2).
- **Minimum document frequency:** 5 (removes extremely rare words appearing in fewer than 0.02% of reviews).
- **Maximum document frequency:** 0.92 (removes near-universal words appearing in more than 92% of reviews, which carry negligible sentiment signal).
- **Term frequency scaling:** Sublinear log scaling (Equation 1).

2.3.2 TF-IDF Configuration for RBF and Polynomial Kernel SVMs

- **Maximum vocabulary size:** 3,500 features, to significantly reduce feature space and computational cost.
- **N-gram range:** Unigrams and bigrams (1, 2).
- **Minimum document frequency:** 5.
- **Maximum document frequency:** 0.92.
- **Term frequency scaling:** Sublinear log scaling.

The resulting feature matrix shapes are shown in Figures 5a and 5b.

For Linear SVM:
For train: Shape of X: (25000, 25000), Shape of y: (25000,)
For test: Shape of X: (25000, 25000), Shape of y: (25000,)

For SVM with RBF and Polynomial kernels:
For train: Shape of X: (5000, 3500), Shape of y: (5000,)
For test: Shape of X: (25000, 3500), Shape of y: (25000,)

(a) Feature matrix shape for Linear SVM.

(b) Feature matrix shape for RBF and Polynomial SVMs.

Figure 5: TF-IDF feature matrix dimensions for Linear SVM (left) and kernel-based SVMs (right).

2.4 Model Selection

Support Vector Machines are chosen as the primary classifier due to their well-established strong performance in high-dimensional feature spaces and their proven effectiveness in text classification tasks [2, 4]. SVMs seek to find the optimal hyperplane that maximizes the margin between classes in the feature space.

Three SVM variants are evaluated in this study:

- **Linear SVM:** Learns a linear decision boundary directly in the TF-IDF feature space.
- **SVM with RBF Kernel:** Maps data into an infinite-dimensional feature space using a Gaussian similarity function; controlled by parameters C and γ .
- **SVM with Polynomial Kernel:** Maps data using a polynomial function of degree d ; controlled by parameters C , γ , and degree d .

2.5 Hyperparameter Tuning and Evaluation

Hyperparameter tuning is performed to study the effect of model complexity on generalization performance. The following hyperparameters are investigated:

- C : Regularization parameter. Controls the tradeoff between maximizing the margin and minimizing classification errors. Smaller C imposes stronger regularization.
- γ : Kernel coefficient for RBF and Polynomial kernels. Controls the reach of influence of individual training samples.
- d : Polynomial degree. Controls the complexity of the decision boundary for the Polynomial kernel.

Four models per kernel type are trained, resulting in 12 models in total:

Linear SVM — $C \in \{0.01, 0.1, 1, 10\}$

SVM with RBF Kernel:

- $C = 0.1, \gamma = 0.001$
- $C = 0.1, \gamma = 0.01$
- $C = 1, \gamma = 0.001$
- $C = 1, \gamma = 0.01$

SVM with Polynomial Kernel:

- $C = 0.1, d = 2$
- $C = 1, d = 2$
- $C = 0.1, d = 3$
- $C = 1, d = 3$

Each model is evaluated using the following metrics: Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and ROC-AUC curve. This combination provides both threshold-dependent and threshold-independent performance assessment.

2.6 Model Validation Strategy

To rigorously assess the generalization behavior of the best-performing model beyond a single fixed train-test evaluation, two complementary validation techniques are employed exclusively for the optimal Linear SVM configuration ($C = 0.1$). Nonlinear kernel variants are excluded from this analysis as they failed to learn meaningful decision boundaries, making generalization analysis irrelevant for those models.

2.6.1 Cross-Validation

5-fold cross-validation is performed directly on the training data vectorized using TF-IDF vectorizer with configurations defined in Section 2.3.1 to verify that the test set performance reported in Section 3.1 is stable and reproducible across different data partitions. The training data is split into 5 equal folds; in each iteration, 4 folds are used for training and the remaining fold is used for validation. This process is repeated 5 times such that every sample serves as a validation sample exactly once. Performance is recorded using Accuracy, Precision, Recall, and F1-Score across all folds, and the mean and standard deviation are reported.

Cross-validation is run sequentially (`n_jobs=1`) rather than in parallel to avoid memory exhaustion on the high-dimensional sparse feature matrix of shape (25000×25000) .

2.6.2 Learning Curve Analysis

A learning curve is constructed by training the model on progressively increasing subsets of training data, ranging from 10% to 80% of the full training set across 10 evenly spaced intervals, and evaluating validation F1-Score at each point using 5-fold cross-validation. The learning curve is a standard diagnostic tool for assessing the bias–variance tradeoff and data sufficiency in supervised learning [7]. The learning curve serves two purposes:

- To determine whether the model has saturated with the available training data or would benefit from additional samples.
- To quantify the training-validation gap and assess the degree of overfitting at the current training set size.

2.6.3 Overfitting Remediation Experiments

Should the learning curve reveal a non-trivial training-validation gap, a systematic set of remediation experiments is conducted by modifying the TF-IDF vectorizer configuration. Two adjustments are investigated independently and in combination:

- **Remedy 1 — Increase `min_df`:** Raising the minimum document frequency threshold from 5 to 10 to remove a larger fraction of rare, potentially noisy vocabulary from the feature space.
- **Remedy 2 — Reduce `max_features`:** Lowering the maximum vocabulary size from 25,000 to 20,000 to constrain the overall feature dimensionality.

For each configuration, the full learning curve experiment is repeated under identical conditions to allow direct comparison of the training-validation gap across all configurations. The results of these experiments are presented in Section 3.5 and analyzed in Section 4.5.

3. Results

3.1 Linear SVM

Four Linear SVM models were trained with $C \in \{0.01, 0.1, 1, 10\}$. The Linear SVM achieved the best overall performance among all evaluated models.

The optimal regularization parameter was found to be $C = 0.1$, yielding an accuracy of 88.72%, an F1-score of 0.887, and an ROC-AUC of 0.956, with a well-balanced precision and recall. Performance improved as C increased from 0.01 to 0.1, and then degraded for larger values of C , clearly demonstrating the bias–variance tradeoff: underfitting at very low C and overfitting at high C .

The complete evaluation metrics for all four Linear SVM configurations are presented in Table 1, with visual comparisons in Figures 6, 7, and 8.

Table 1: Accuracy, Precision, Recall, F1-Score, and AUC for four Linear SVM models with different values of regularization parameter C .

Model	C	Accuracy (%)	Precision	Recall	F1-Score	AUC
Linear SVM	0.01	86.520	0.8519	0.8841	0.8677	0.9406
Linear SVM	0.1	88.720	0.8818	0.8942	0.8880	0.9563
Linear SVM	1.0	87.672	0.8794	0.8732	0.8763	0.9491
Linear SVM	10.0	85.444	0.8596	0.8473	0.8534	0.9327

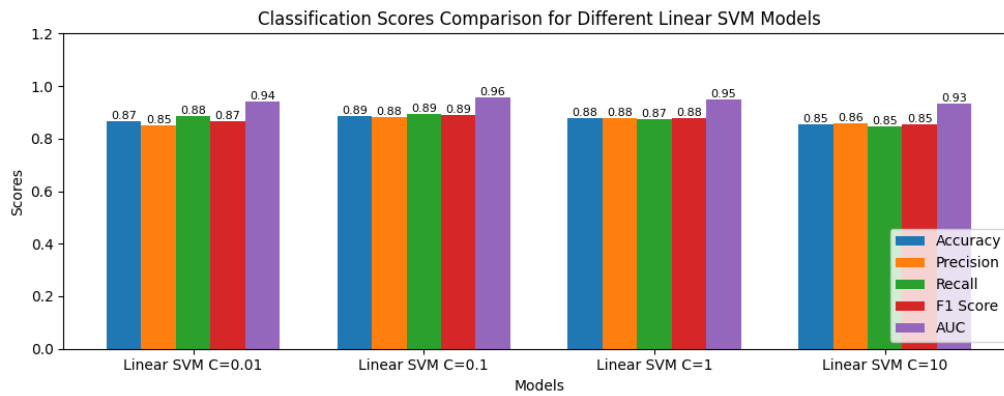


Figure 6: Classification score comparison across four Linear SVM models with different C values.

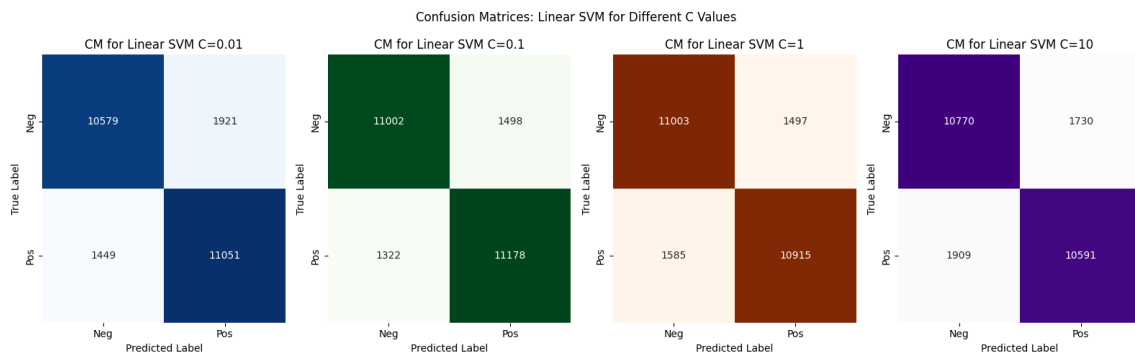


Figure 7: Confusion matrices for four Linear SVM models with different values of C .

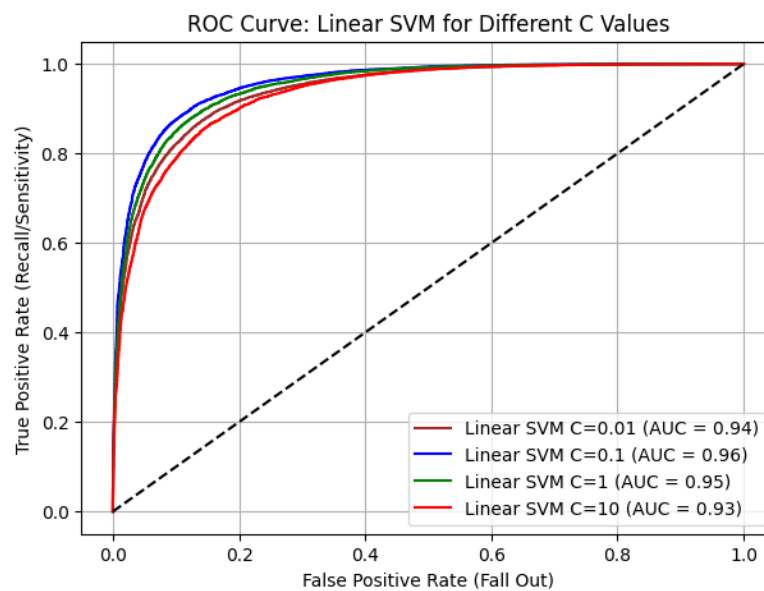


Figure 8: ROC curves for four Linear SVM models. The model at $C = 0.1$ achieves the highest AUC of 0.956.

3.2 SVM with RBF Kernel

Four SVM models with RBF kernel were trained on a reduced subset of 5,000 training samples due to high computational cost. All four configurations produced nearly identical results regardless of the specific values of C and γ tested.

The models achieved an accuracy close to random guessing ($\approx 51\%$), with very high recall (0.996) but extremely low precision (0.503). ROC-AUC values remained stagnant at approximately 0.58, indicating near-chance class separation. The near-perfect recall combined with near-random precision is a strong indicator that the models are collapsing to a trivial strategy of predicting the positive class for virtually every sample. This is clearly corroborated by the confusion matrices in Figure 10.

The complete evaluation metrics are presented in Table 2.

Table 2: Accuracy, Precision, Recall, F1-Score, and AUC for four SVM models with RBF kernel across different C and γ values.

Model	C	γ	Acc. (%)	Precision	Recall	F1
RBF SVM	0.1	0.001	50.648	0.5033	0.9963	0.6687
RBF SVM	0.1	0.010	50.648	0.5033	0.9963	0.6687
RBF SVM	1.0	0.001	50.648	0.5033	0.9963	0.6687
RBF SVM	1.0	0.010	50.648	0.5033	0.9963	0.6687

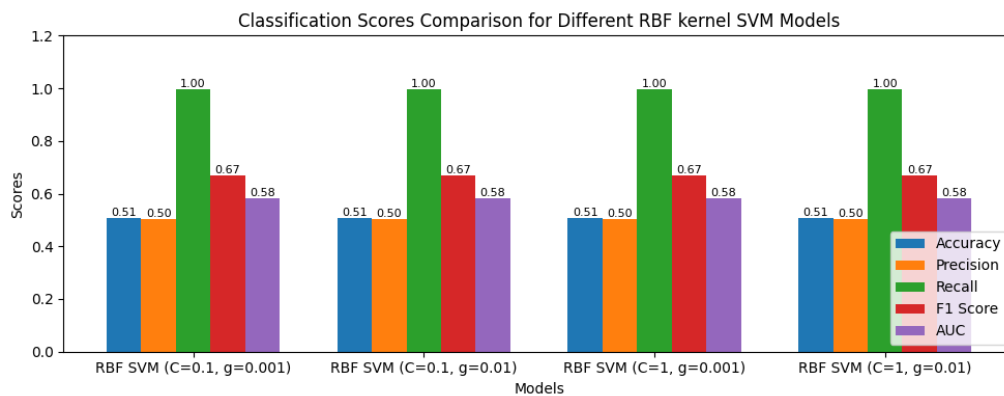


Figure 9: Classification score comparison across four SVM models with RBF kernel.

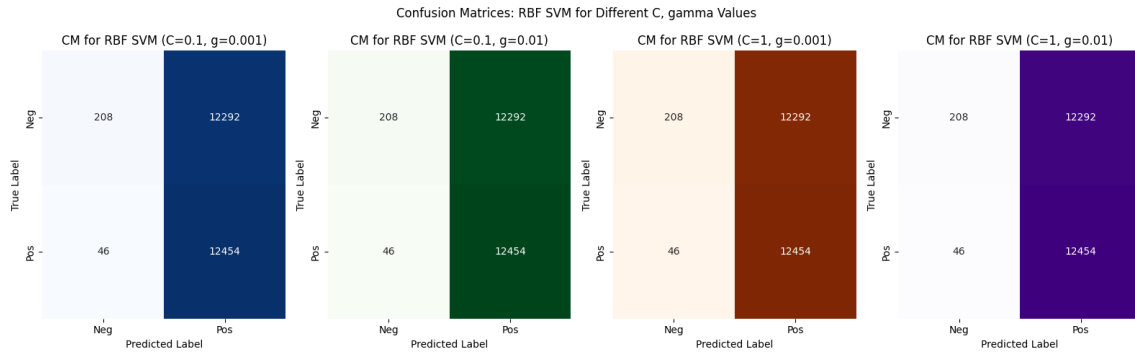


Figure 10: Confusion matrices for four RBF kernel SVM models, confirming that models predict almost exclusively the positive class.

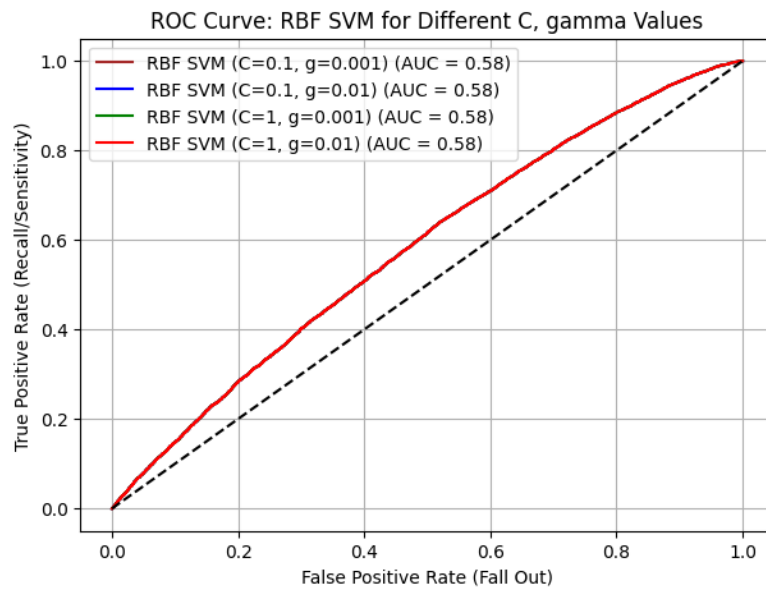


Figure 11: ROC curves for four RBF kernel SVM models ($AUC \approx 0.58$ for all configurations).

3.3 SVM with Polynomial Kernel

Four SVM models with Polynomial kernel were trained under the same reduced-data regime as the RBF experiments. Similar to the RBF case, all Polynomial SVM configurations exhibited near-random performance across all tested values of C and degree d .

Accuracy remained near 50%, and ROC-AUC values were approximately 0.56. As with the RBF kernel models, high recall combined with very low precision indicates a collapse to predominantly positive-class predictions, as confirmed by the confusion matrices in Figure 13. The evaluation metrics are presented in Table 3.

Table 3: Accuracy, Precision, Recall, F1-Score, and AUC for four SVM models with Polynomial kernel across different C and degree values.

Model	C	Degree	Acc. (%)	Precision	Recall	F1
Poly SVM	0.1	2	50.000	0.5000	1.0000	0.6667
Poly SVM	0.1	3	50.000	0.5000	1.0000	0.6667
Poly SVM	1.0	2	53.596	0.5237	0.7947	0.6314
Poly SVM	1.0	3	50.784	0.5040	0.9918	0.6684

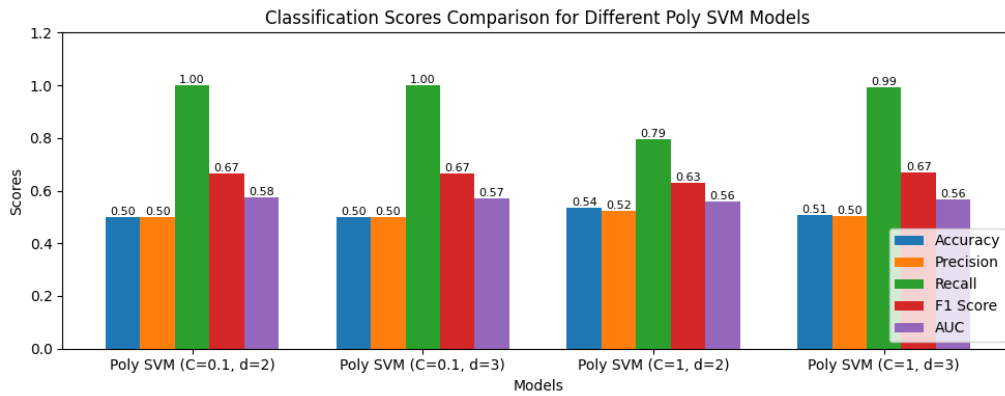


Figure 12: Classification score comparison across four SVM models with Polynomial kernel.

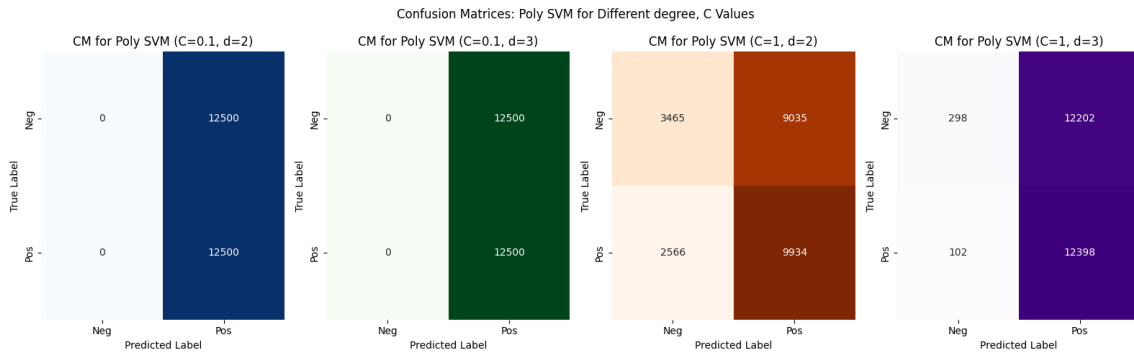


Figure 13: Confusion matrices for four Polynomial kernel SVM models.

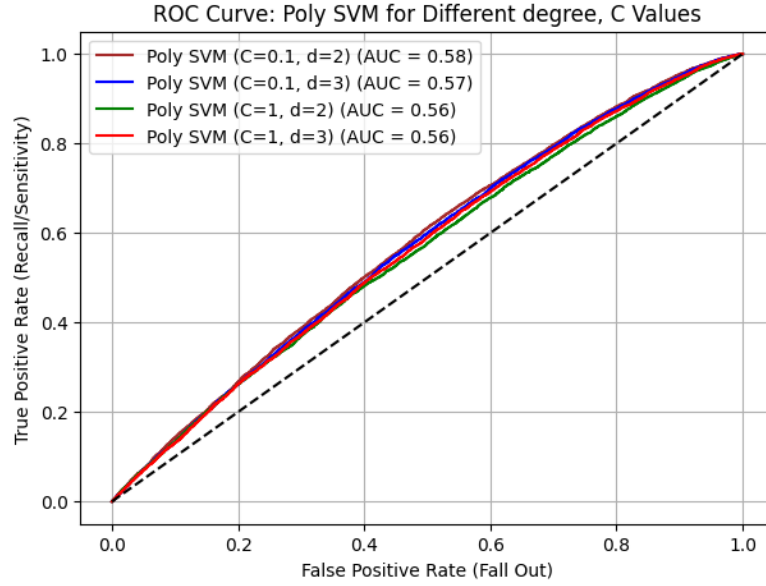


Figure 14: ROC curves for four Polynomial kernel SVM models ($AUC \approx 0.56$ for all configurations).

3.4 Final Comparison Across All Models

A consolidated comparison across all 12 trained models (4 Linear, 4 RBF, 4 Polynomial) is presented in Table 4 and Figure 15.

The results unambiguously demonstrate that the Linear SVM significantly outperforms both nonlinear kernel variants across every evaluation metric. The best-performing model overall is Linear SVM with $C = 0.1$, achieving an accuracy of 88.72%, F1-score of 0.887, and AUC of 0.956, with a well-balanced precision (0.882) and recall (0.894).

Table 4: Consolidated evaluation results for all 12 SVM models.

SL.	Model	C	γ	Degree	Acc. (%)	Precision	Recall	F1
0	Linear SVM	0.01	N/A	N/A	86.520	0.8519	0.8841	0.8677
1	Linear SVM	0.10	N/A	N/A	88.720	0.8818	0.8942	0.8880
2	Linear SVM	1.00	N/A	N/A	87.672	0.8794	0.8732	0.8763
3	Linear SVM	10.00	N/A	N/A	85.444	0.8596	0.8473	0.8534
4	RBF SVM	0.1	0.001	N/A	50.648	0.5033	0.9963	0.6687
5	RBF SVM	0.1	0.010	N/A	50.648	0.5033	0.9963	0.6687
6	RBF SVM	1.0	0.001	N/A	50.648	0.5033	0.9963	0.6687
7	RBF SVM	1.0	0.010	N/A	50.648	0.5033	0.9963	0.6687
8	Poly SVM	0.1	N/A	2	50.000	0.5000	1.0000	0.6667
9	Poly SVM	0.1	N/A	3	50.000	0.5000	1.0000	0.6667
10	Poly SVM	1.0	N/A	2	53.596	0.5237	0.7947	0.6314
11	Poly SVM	1.0	N/A	3	50.784	0.5040	0.9918	0.6684

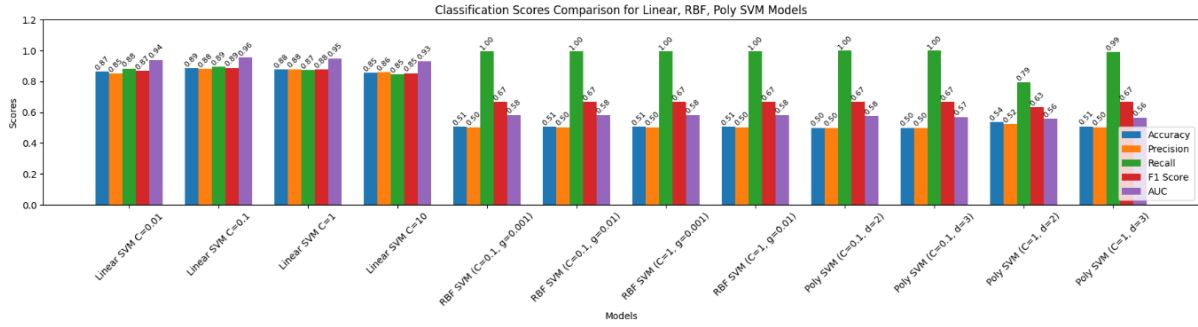


Figure 15: Classification score comparison across all 12 SVM models (Linear, RBF, and Polynomial).

3.5 Model Stability and Generalization

3.5.1 Cross-Validation Results

5-fold cross-validation performed on training data vectorized using TFIDF configuration defined in Section 2.3.1 using the best Linear SVM model ($C = 0.1$) yields the results presented in Table 5 and Table 6, with per-fold scores visualized in Figure 16.

Table 5: Per-fold Accuracy, Precision, Recall, and F1-Score for the best Linear SVM ($C = 0.1$) across all 5 cross-validation folds.

Fold	Accuracy (%)	Precision	Recall	F1 Score
Fold 1	88.82	0.8793	0.9000	0.8895
Fold 2	88.58	0.8751	0.9000	0.8874
Fold 3	89.26	0.8791	0.9104	0.8945
Fold 4	89.02	0.8730	0.9132	0.8927
Fold 5	89.62	0.8883	0.9064	0.8972

Table 6: 5-fold cross-validation results for the best Linear SVM ($C = 0.1$).

Metric	Mean	Std Dev
Accuracy	0.8906	0.0036
Precision	0.8790	0.0052
Recall	0.9060	0.0054
F1 Score	0.8923	0.0035

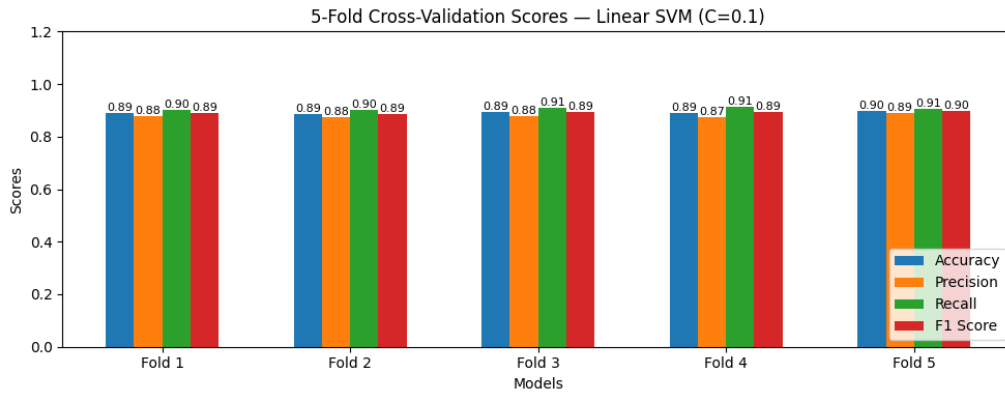


Figure 16: Per-fold Accuracy, Precision, Recall, and F1-Score for the best Linear SVM ($C = 0.1$) across all 5 cross-validation folds.

All four metrics remain consistent across all 5 folds, with standard deviations not exceeding 0.0054. The mean CV Accuracy of 89.06% and mean F1-Score of 0.8923 are marginally higher than the single fixed test split results reported in Table 1 (88.72%, 0.887), confirming that those results were not optimistic outliers produced by a favorable data partition.

3.5.2 Learning Curve Results

The learning curve for the best Linear SVM ($C = 0.1$), constructed across 10 training size intervals from 10% to 80% of the full training set, is shown in Figure 17. Quantitative results at the largest training size evaluated are summarized below.

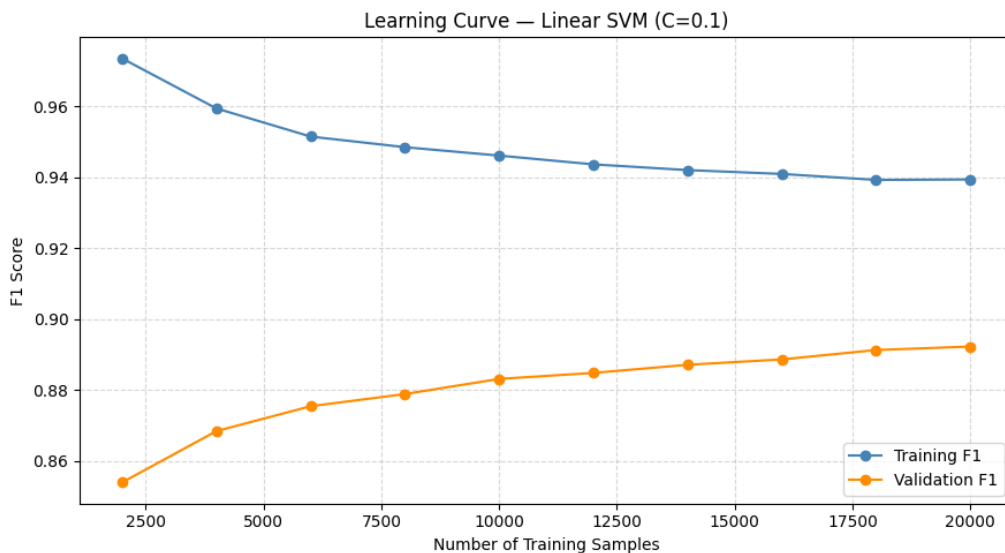


Figure 17: Learning curve for the best Linear SVM ($C = 0.1$). Training F1 decreases as sample size grows while Validation F1 rises steadily, with neither curve plateauing at the maximum evaluated training size.

- **Training F1 at full size:** 0.9394
- **Validation F1 at full size:** 0.8923
- **Training-Validation Gap:** 0.0472

The validation F1 curve has not plateaued at the maximum evaluated training size of approximately 20,000 samples, indicating that the model has not yet saturated with the available data.

3.5.3 Overfitting Remediation Results

The remediation experiments described in Section 2.6 are carried out across four TF-IDF configurations. The resulting training-validation gaps at full training size are presented in Table 7, with the corresponding validation F1 learning curves shown in Figure 18.

Table 7: Training F1, Validation F1, and training-validation gap at full training size for the baseline and all three remediation configurations.

Variant	max_f	min_df	Train F1	Val F1	Gap
Baseline	25000	5	0.9394	0.8923	0.0472
Remedy 1	25000	10	0.9393	0.8920	0.0473
Remedy 2	20000	5	0.9374	0.8923	0.0451
Remedy 1+2	20000	10	0.9373	0.8920	0.0453

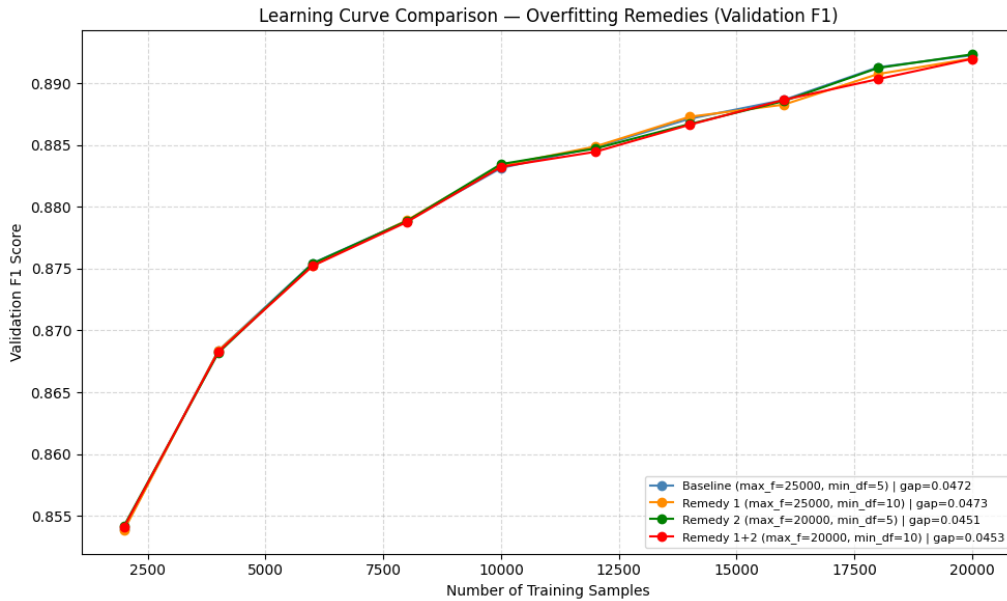


Figure 18: Validation F1 learning curves for the baseline and all three remediation configurations. All four curves are nearly indistinguishable across the full range of training sizes.

All four configurations yield nearly identical validation F1 curves and training-validation gaps, with a maximum difference of 0.002 between the best and worst configurations. Neither remedy, individually nor in combination, produced a meaningful reduction in the overfitting gap. The baseline configuration (`max_features=25000`, `min_df=5`) is therefore retained as the optimal TF-IDF setting. The interpretation of these findings is presented in Section 4.5.

4. Discussion

4.1 Why Linear SVM Performed Best

The Linear SVM achieved the best performance among all evaluated models. This outcome is well-explained by the intrinsic properties of TF-IDF representations. TF-IDF feature vectors are both high-dimensional and sparse: each document is represented as a vector of thousands of features, the vast majority of which are zero for any given review. In such high-dimensional sparse spaces, positive and negative sentiment classes tend to become approximately linearly separable [5]. The Linear SVM exploits this property directly by learning a stable linear decision boundary that generalizes well to unseen data, while remaining computationally efficient on the full 25,000-sample training set.

4.2 Bias–Variance Tradeoff in Linear SVM

The effect of the regularization parameter C in Linear SVM clearly demonstrates the classical bias–variance tradeoff. At very small values of C (e.g., $C = 0.01$), the model is heavily regularized, constraining the decision boundary and preventing the model from capturing all sentiment-relevant patterns; this leads to underfitting (high bias). As C increases toward 0.1, the model gains sufficient flexibility to learn the data well, reaching the optimal balance point with the highest F1-score and AUC. Beyond $C = 0.1$, further relaxation of regularization results in overfitting (high variance), causing the model to fit training noise and lose generalization, as reflected in the declining test performance at $C = 1$ and $C = 10$.

4.3 Why RBF and Polynomial Kernels Failed

SVMs with RBF and Polynomial kernels performed dramatically worse than the Linear SVM in this study. These kernel functions rely on distance-based similarity measures or complex polynomial feature interactions in the implicit kernel space. However, such distance-based measures become fundamentally unreliable in the high-dimensional sparse setting of TF-IDF vectors, a phenomenon often referred to as the *curse of dimensionality* [6]. In very high-dimensional sparse spaces, pairwise Euclidean distances between samples converge, rendering the locality assumptions implicit in the RBF kernel ineffective. As a result, both RBF and Polynomial SVMs struggled to form a meaningful decision boundary and collapsed into predicting a single class (positive) for virtually all test samples, performing no better than random guessing. This is confirmed by the near-diagonal ROC curves in Figures 11 and 14.

4.4 Misclassification Analysis

To better understand the limitations of the best-performing model (Linear SVM, $C = 0.1$), a systematic error analysis was conducted on the misclassified samples from the test set. The analysis reveals the following recurring patterns:

1. **Long reviews with mixed sentiments.** Many misclassified reviews are lengthy and contain interleaved positive and negative expressions (e.g., “*I didn’t expect much... some parts were boring... but overall it was surprisingly enjoyable.*”). This mixed sentiment creates conflicting feature signals that are difficult to resolve with a single linear decision boundary.
2. **Negation and contrast structures.** TF-IDF represents text as a bag of independent weighted tokens and therefore cannot capture the scope of negation or contrast markers

such as “not,” “but,” and “however.” For example, the phrase “*didn’t expect it to be good, but...*” causes the model to assign a positive weight to the word “good” even in a negated context.

3. **Ironic and mildly opinionated language.** Some reviews employ subtle or ironic phrasing (e.g., “*It’s not the worst thing you’ll ever see*”, “*I suppose it has its moments*”), which lacks the strong sentiment keywords that a TF-IDF-based model relies upon. Ambiguous feature representations in such cases make correct label assignment difficult.
4. **Rare vocabulary filtered by TF-IDF.** The minimum document frequency threshold of 5 causes rare but potentially sentiment-bearing words to be excluded from the feature space, resulting in information loss for reviews containing uncommon vocabulary.
5. **Very short reviews.** Extremely short reviews (e.g., “*Disappointing. Expected more.*”) provide insufficient textual evidence for reliable sentiment classification.

Notably, misclassifications occurred across a wide range of review lengths and types, indicating that errors are not strongly correlated with review length per se but rather with linguistic complexity, contextual structure, and the presence of ambiguous or implicit sentiment.

4.5 Generalization Analysis and Overfitting

4.5.1 Model Stability Confirmed by Cross-Validation

The 5-fold cross-validation results presented in Table 5 and Table 6 provide strong evidence that the best Linear SVM ($C = 0.1$) is a stable and reliable classifier. The standard deviation across all four metrics does not exceed 0.0054, and per-fold accuracy ranges only between 88.58% and 89.62% — a spread of just 1.04 percentage points across all five independently constructed validation sets.

This finding carries an important implication: the strong test performance reported in Section 3.1 is not a statistical artifact of a particularly favorable train-test split. The model generalizes consistently regardless of how the training data is partitioned, which is a necessary condition for trusting the reported metrics in a real-world deployment context.

Furthermore, the mean CV Accuracy (89.06%) and mean CV F1-Score (0.8923) are marginally higher than the single fixed test split results (88.72%, 0.887). This slight upward shift is expected — cross-validation averages over multiple complementary partitions of the training data and therefore provides a less pessimistic estimate of true generalization performance than any single split.

4.5.2 Interpretation of the Learning Curve

The learning curve in Figure 17 reveals two distinct and meaningful behavioral patterns that together characterize the current state of the model.

The training F1 begins at ≈ 0.975 at the smallest training size and decreases monotonically as more samples are added, converging toward 0.9394 at full size. This is the expected behavior of a well-functioning model — as the training set grows more diverse, perfect memorization becomes harder and the training score naturally approaches the true generalization level.

The validation F1 begins at ≈ 0.855 and rises consistently throughout, reaching 0.8923 at approximately 20,000 samples with no sign of plateauing. This continued upward trajectory is a direct empirical signal that the model has not yet saturated with the available data — additional

training samples beyond 25,000 would likely yield further improvement in generalization performance.

The training-validation gap of 0.0472 at full training size indicates moderate overfitting. However, it is important to distinguish between two components that contribute to this gap [7]:

- **Data-reducible component:** The portion of the gap attributable to insufficient training data. Since the validation curve has not plateaued, more data would partially close this gap.
- **Model-irreducible component:** The portion attributable to the fundamental representational limitations of TF-IDF and Linear SVM — phenomena such as negation, mixed sentiment, and implicit irony that the model structurally cannot capture regardless of training set size or vocabulary configuration.

The remediation experiments described below allow us to empirically determine which of these two components dominates the observed gap.

4.5.3 Why the Overfitting Remedies Failed

The remediation experiments presented in Table 7 and Figure 18 produce a clear and conclusive finding: neither increasing `min_df`, nor reducing `max_features`, nor applying both adjustments simultaneously produced a meaningful reduction in the training-validation gap. The maximum difference in gap across all four configurations is just 0.002 — well within the range of random variation.

This result is analytically significant. Both remedies specifically target the vocabulary composition of the TF-IDF feature space — one by removing rare words, the other by capping total vocabulary size. The fact that neither intervention affected the gap in any meaningful way allows us to draw the following conclusion with confidence: *the overfitting gap is not caused by noisy or over-specific vocabulary in the feature space.*

The deeper explanation lies in the nature of TF-IDF itself. Regardless of vocabulary size or document frequency thresholds, TF-IDF always produces a sparse bag-of-words representation in which each token is treated as an independent, context-free feature. The model therefore has no mechanism to represent:

- The scope of negation (e.g., “not good” vs. “good”).
- Contrast structures (e.g., “boring at first, but ultimately rewarding”).
- Ironical or implicit sentiment (e.g., “I suppose it has its moments”).
- Long-range semantic dependencies between distant parts of a review.

These are precisely the linguistic phenomena identified in the misclassification analysis of Section 4.4 as the primary sources of model error. The convergence of the misclassification analysis and the remediation experiments onto the same root cause — the representational ceiling of bag-of-words features — constitutes a coherent and mutually reinforcing analytical finding.

The baseline TF-IDF configuration (`max_features=25000`, `min_df=5`) is therefore retained as the optimal setting. Meaningfully reducing the overfitting gap will require replacing the TF-IDF representation entirely with contextually aware feature representations, such as those produced by transformer-based language models like BERT [8], which encode token meaning as a function of surrounding context rather than as an isolated frequency statistic.

4.6 Limitations and Future Work

The present study is limited to classical machine learning methods employing TF-IDF features. As demonstrated empirically in Section 4.5, the overfitting gap observed in the best-performing model is irreducible through vocabulary-level adjustments alone, confirming that the ceiling of this approach is fundamentally representational rather than a matter of tuning. TF-IDF cannot capture contextual or semantic relationships between words beyond raw term frequency statistics — a structural limitation that no amount of hyperparameter adjustment can overcome within the bag-of-words paradigm.

Future work may address these limitations through:

- **Transformer-based models** (e.g., BERT [8]), which model deep bidirectional contextual relationships and have demonstrated state-of-the-art performance on sentiment analysis benchmarks.
- **Word embeddings** (e.g., Word2Vec, GloVe, FastText) as alternative feature representations that encode semantic similarity between tokens, addressing the context-blindness of TF-IDF.
- **Hybrid architectures** combining classical SVM classifiers with dense embedding-based features to retain the efficiency of SVMs while gaining richer semantic representations.
- **Enhanced preprocessing** targeting negation handling and contrast structure identification, which may partially mitigate the linguistic phenomena identified in Section 4.4 as primary sources of misclassification.

5. Conclusion

In this study, I presented a systematic empirical comparison of three SVM kernel variants — Linear, RBF, and Polynomial — for binary sentiment classification on the IMDB movie review dataset using TF-IDF feature representations, with additional analysis of model stability, generalization behavior, and the fundamental limits of the bag-of-words paradigm.

The key findings are as follows. The Linear SVM significantly outperforms both nonlinear kernel variants across all evaluation metrics, achieving an accuracy of 88.72%, F1-score of 0.887, and AUC of 0.956 at the optimal regularization parameter $C = 0.1$. The failure of RBF and Polynomial kernels is attributed to the fundamental incompatibility of distance-based kernel similarity measures with high-dimensional sparse TF-IDF representations. Misclassification analysis reveals that the primary error-inducing phenomena are mixed sentiment, negation, implicit irony, and rare vocabulary — linguistic properties that lie beyond the representational capacity of TF-IDF and linear classifiers.

5-fold cross-validation confirms that the best model achieves a stable CV accuracy of 89.06% $\pm$ 0.0036 across all folds, establishing that the reported results are reproducible and not an artifact of a favorable data partition. Learning curve analysis reveals a moderate training-validation gap of 0.047, and systematic remediation experiments demonstrate conclusively that this gap is irreducible through TF-IDF vocabulary adjustments alone — converging with the misclassification analysis to identify the representational ceiling of bag-of-words features as the fundamental limiting factor of this approach.

These findings are consistent with prior literature and reaffirm that Linear SVMs with TF-IDF features remain a strong, computationally efficient baseline for text classification. However, both the misclassification analysis and the generalization experiments point to the same conclusion:

meaningfully surpassing the performance ceiling demonstrated here will require models capable of genuine contextual language understanding, such as transformer-based architectures.

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A. Supplementary Materials

The following supplementary materials accompany this report:

- **Misclassification Report (CSV):** Contains detailed records of all misclassified samples produced by the best-performing Linear SVM model ($C = 0.1$), including actual and predicted labels, raw and preprocessed review text, and basic text statistics (character count, word count, sentence count).
[Download Misclassification Report \(CSV\)](#)
- **Implementation Notebook (Jupyter Notebook):** Contains complete implementation of all pipeline stages — data preprocessing, TF-IDF feature extraction, model training, hyperparameter tuning, evaluation, comparison, and visualization — as used in this study.
[Download Implementation Notebook \(.ipynb\)](#)